higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

13 July 2022 (X-paper)
09:00–12:00

Calculators may not be used. Tekeninstrumente mag gebruik word.

This question paper consists of 7 pages.

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELCTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a new page.
5. Use only black or a blue pen.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 $31_8 + 3_{16} =$

A 10000_2
B 36_{10}
C 11000_2
D 28_{10}

1.2 $10,4_8 + 10,5_{10} =$

A $28,6_8$
B 19
C 19,25
D $26,4_8$

1.3 $13_{16} – 1010,1_2 =$

A $10,10_{16}$
B $1000,1_{16}$
C $1000,1_8$
D $1000,1_2$

1.4 $4_8 \times 4_{10} =$

A 16_{10}
B 16_8
C 16_{16}
D 1000_2

1.5 $8_{16} \div 100_2 =$

A 11_2
B $100,11_2$
C $100,01_2$
D 10_2

(5 × 2) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (2.1–2.8) in the ANSWER BOOK.

2.1 $A+1+0=A$

2.2 $A+\overline{A}+1=A$

2.3 $A+\overline{A}+0=A$

2.4 $\overline{A}\bullet A\bullet 1=A$

2.5 $A\bullet\overline{A}\bullet 1=1$

2.6 $A\bullet\overline{A}\bullet 1=0$

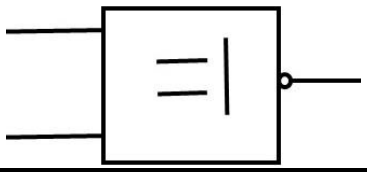
2.7 $A+(\overline{A}\bullet\overline{B})=A+\overline{B}$

2.8 $\overline{A}\bullet(\overline{\overline{A}}+\overline{B})=\overline{A}$

(8 × 1) **[8]**

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (3.1–3.6) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	This logic gate will have a binary 0 on the output when all the inputs are binary 1's	A	7
3.2	This code is a weighed self-inverting code	B	Octal
3.3	This numbering system makes use of the following symbols. 0,1,2,3,4,5,6,7	C	Exclusive or gate
3.4	This is the maximum count for a 3-bit asynchronous binary counter	D	NAND gate
3.5	<table><tr><th>A</th><th>B</th><th>OUTPUT</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	E	Hexadecimal
3.6	The base of this numbering system is 16	F	Exclusive NOR gate
3.7		G	4221 BCD

(7 × 1) **[7]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

Complete the following calculation in the binary number system. All steps must be indicated clearly.

4.1 $1011{,}011_2 - 11000{,}0101_2$ (Make use of the 1's complement method)

4.2 $1001{,}11_2 - 1110{,}011_2$ (Make use of the 2's complement method)

4.3 $10110{,}01_2 \times 1001{,}01_2$

4.4 $101010{,}100111_2 \div 11{,}011_2$

(4 × 5) **[20]**

QUESTION 5

Draw the circuit diagrams of the Boolean expressions below. Do not change or simplify the expressions:

5.1 $\overline{A}\overline{B} + B\overline{C} + \overline{A}B\overline{C} = f$

5.2 $\overline{(\overline{A+B})(A+\overline{B}+\overline{C})(\overline{A}+C)} = f$

5.3 $\overline{\overline{AB} \oplus \overline{B+\overline{C}}} + \overline{B \oplus C} = f$

(3 × 5) **[15]**

QUESTION 6

Simplify the following Boolean expressions:

6.1 $(\overline{A}C + B)(B + \overline{C})(AB + C)(A + \overline{B}) = f$

6.2 $\overline{(\overline{A}BC + A\overline{B}C)}\,\overline{(ABC + AB\overline{C})} = f$

(2 × 5) **[10]**

QUESTION 7

7.1 Give the symbol and truth table for each of the following flip-flops:

7.1.1 SR

7.1.2 D-TYPE

7.1.3 JK

(3 × 4) (12)

7.2 Draw the circuit diagram of a four-bit series shift right register using D flip-flops with a serial input and parallel outputs. Show all control and input/output points. (10)

7.3 Draw a circuit diagram of a half adder making use of a AND gate and an EXCLUSIVE OR-gate. Label the inputs and outputs clearly. Also give the truth table of the circuit diagram. (8)

[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100